Hugo Marquerie 11/03/2025

Lema de Borel-Cantelli II

Lema 1 (de Borel-Cantelli II). *Sean* $(A_n)_{n\in\mathbb{N}} \subset \Sigma$ *independientes en* $(\Omega, \Sigma, \mathbb{P})$

$$\sum_{n\in\mathbb{N}} \mathbb{P}(A_n) = \infty \implies \mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 1.$$

***Demostración*:** Sean $M < N < \infty$, entonces

$$\mathbb{P}\left(\bigcap_{n=M}^{N} A_n^c\right) = \prod_{n=M}^{N} \mathbb{P}(A_n^c) = \prod_{n=M}^{N} (1 - \mathbb{P}(A_n)) \leqslant \prod_{n=M}^{N} e^{-\mathbb{P}(A_n)} \text{ porque } 1 - x \leqslant e^{-x}$$
$$\leqslant e^{-\sum_{n=M}^{N} \mathbb{P}(A_n)} \xrightarrow{N\to\infty} 0, \quad \forall M < \infty.$$

Como $\Omega = \bigcup_{n=M}^{\infty} A_n \sqcup \bigcap_{n=M}^{\infty} A_n^c$, hemos visto que $\mathbb{P}\left(\bigcap_{n=M}^{\infty} A_n^c\right) = \lim_{N\to\infty} \mathbb{P}\left(\bigcap_{n=M}^{N} A_n^c\right) = 0.$

$$\implies \forall M < \infty : \mathbb{P}\left(\bigcup_{n=M}^{\infty} A_n\right) = 1 \implies \mathbb{P}\left(\bigcap_{M=1}^{\infty} \bigcup_{n=M}^{\infty} A_n\right) = 1.$$

Entonces, concluimos que $\mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 1.$ ■

Referenciado en

- `quijote-infinito`
- `ley-fuerte-grandes-numeros`

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